CHAPTER 17

Calibrating to Swaptions

17.1 Why swaptions are the harder half

Chapter 7 flagged the core difficulty: a swap rate $S(t)$ is a weighted average of many individual forward rates, with random weights, so its volatility is not a primitive BGM input but something that must be *derived* from the volatilities and correlations of every forward rate underlying it. Calibrating to the swaption market therefore means solving an inverse problem: given a target swaption volatility, find volatility and (especially) correlation parameters such that the swap rate volatility the model *implies* matches it.

17.2 Approximating the swap rate's volatility

The swap rate has no exact closed-form volatility in terms of the individual σ_i and ρ_{ij}, because $S(t)$ is a genuinely nonlinear function of the $L_i(t)$'s (through the annuity, which is itself a function of all of them). The standard practitioner approximation, usually attributed to Rebonato, linearizes this relationship by freezing the weights at their time-0 values.

Key result: Rebonato's swaption approximation

Write $S(t) \approx \sum_i w_i(0)\, L_i(t)$ for weights $w_i(0) = \partial S/\partial L_i$ evaluated at today's curve (a computable, purely deterministic quantity built from today's discount factors and annuity — Chapter 4's machinery). *Freezing* these weights at their time-0 values (rather than letting them evolve stochastically, which is what makes the exact problem hard) and matching the resulting variance of $S(T)$ to Black's implied variance for the swaption gives

$$(\sigma^{\text{swaption}})^2\, T \;\approx\; \frac{1}{S(0)^2} \sum_{i,j} w_i(0)\, w_j(0)\, L_i(0)\, L_j(0)\, \rho_{ij} \int_0^T \sigma_i(t)\, \sigma_j(t)\, \mathrm{d}t$$

where the sum runs over every forward rate underlying the swap, T is the swaption expiry, and σ^{swaption} is the (lognormal) Black implied volatility. This single formula is the bridge between BGM's per-rate parameters and the market's swaption quotes, and it is what a calibration engine actually evaluates, repeatedly, inside an optimizer loop.

Read the formula as: swap rate variance is a weighted sum of pairwise forward-rate covariances, where the weights come from how sensitive the swap rate is to each underlying forward rate. The diagonal terms ($i = j$) contribute each rate's own variance, weighted by how much that rate matters to the swap; the off-diagonal terms ($i \neq j$) contribute covariance, and it is precisely these cross-terms — entirely absent from a caplet calibration, which only ever sees one rate at a time — that make the correlation matrix ρ_{ij} visible to the market for the first time in this book. **This is the single most important fact in this chapter**: caplet prices alone do not pin down correlation at all; only the swaption market does.

Worked example 17.1 — a two-rate swaption approximation.
A 1y $\times$ 1y swaption (so the underlying swap has just two semi-annual periods, $i = 1, 2$, with the option expiring at $T = 1$) has weights $w_1(0) = 0.51$, $w_2(0) = 0.49$ (roughly equal, as expected for a short 2-period swap), forward rates $L_1(0) = L_2(0) = 4.05\%$, swap rate $S(0) = 4.05\%$, instantaneous vols (assumed constant over $[0, 1]$ for simplicity) $\sigma_1 = 21\%, \sigma_2 = 23\%$, and correlation $\rho_{12} = 0.92$. Find the approximate swaption Black vol.
Step 1 — the three pairwise terms (each $\int_0^1 \sigma_i\sigma_j\,dt = \sigma_i\sigma_j$ since constant vols over unit time).

$$(1,1): \; w_1^2L_1^2\sigma_1^2 = 0.51^2 \times 0.0405^2 \times 0.21^2 = 1.881 \times 10^{-5}$$

$$(2,2): \; w_2^2L_2^2\sigma_2^2 = 0.49^2 \times 0.0405^2 \times 0.23^2 = 2.083 \times 10^{-5}$$

$$(1,2) + (2,1): \; 2\,w_1w_2L_1L_2\rho_{12}\sigma_1\sigma_2 = 2 \times 0.51 \times 0.49 \times 0.0405^2 \times 0.92 \times 0.21 \times 0.23 = 3.643 \times 10^{-5}$$

Step 2 — sum and normalise by $S(0)^2$.

$$\text{Sum} = 1.881 + 2.083 + 3.643 = 7.607\ (\times 10^{-5}), \qquad S(0)^2 = 0.0405^2 = 1.640 \times 10^{-3}$$

$$(\sigma^{\text{swaption}})^2 \times 1 \approx \frac{7.607 \times 10^{-5}}{1.640 \times 10^{-3}} = 0.04638$$

Step 3 — take the square root.

$$\sigma^{\text{swaption}} \approx \sqrt{0.04638} = 0.2154 \;\rightarrow\; 21.5\%$$

Sanity check. 21.5% sits, as it should, between the two input per-rate volatilities (21% and 23%) — with two rates carrying almost equal weight, almost equal level, and a very high correlation ($\rho = 0.92$), the swap rate should behave almost like a single blended rate with a volatility close to a weighted average of the two inputs, and 21.5% is exactly that: barely above $\sigma_1 = 21\%$, comfortably below $\sigma_2 = 23\%$.
Interpretation. With only two underlying rates, both volatility and correlation visibly move the answer; with a real 20-period swap, the same formula — now a 20×20 double sum — is exactly what an optimizer evaluates on every iteration while searching for the correlation and volatility parameters that best match an entire grid of quoted swaption vols simultaneously. Note also how close the result sits to a simple weighted average of σ_1, σ_2: this is a direct consequence of the high assumed correlation ($\rho = 0.92$) here. Re-running the same numbers with $\rho = 0.40$ instead would pull the result down substantially below either input — diversification between imperfectly-correlated rates genuinely lowers the swap rate's volatility, exactly as it lowers the volatility of any imperfectly-correlated portfolio.

17.3 The terminal correlation problem

PITFALL

The correlation ρ_{ij} appearing in this chapter's formula is the *instantaneous* correlation between two forward rates' driving Brownian motions — a primitive model input. What actually governs a swaption's price, however, is closer to the **terminal correlation** between the rates' realised values at expiry, which is a more complex, model-implied quantity that depends on the whole path of instantaneous correlations and volatilities over $[0, T]$, not just their values at a single instant. The frozen-weights formula above is itself already an approximation to this deeper problem, accurate for typical, realistic market parameters but progressively less accurate for long-dated swaptions on long underlying swaps, where the frozen-weight assumption has more room to drift from reality over the life of the option. A calibration engine that fits beautifully to short-expiry, short-tenor swaptions and then shows unexplained residual error specifically on the long-dated corners of the cube is very often exhibiting exactly this approximation breaking down, not a genuinely bad correlation fit.

ON THE DESK

When a swaption calibration residual is unexpectedly large for one specific point on the cube, an experienced quant's first move is rarely to blame the optimizer. The more common causes, roughly in order of likelihood: a stale or bad input quote at that specific expiry/tenor; the frozen-weights approximation genuinely breaking down at that point on the cube (long expiry, long tenor); or a correlation parametric form (Chapter 14) too rigid to simultaneously fit both the caplet-implied short-end structure and this specific swaption's requirements. Diagnosing which of the three is happening — rather than simply loosening tolerances or throwing more optimizer iterations at the problem — is a genuine, recurring skill on a calibration desk, and Chapter 20 returns to it as a first-class operational concern.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. State Rebonato's frozen-weights swaption approximation and explain what each factor (weight, forward level, correlation, volatility) does in the formula.
2. Explain why caplet-only calibration cannot pin down correlation, and why the swaption market is the first place correlation becomes visible.
3. Explain, at a conceptual level, why the frozen-weights approximation degrades for long-expiry, long-tenor swaptions, and what "terminal correlation" means as distinct from instantaneous correlation.